10-07 第2篇

阅读理解（说明文） Why Do Cats Purr?

① Walk quietly past a sleeping cat. You may hear a soft, steady sound, something between a little engine and a distant boat: **a cat purring**. Almost every cat owner has felt that warm buzz under their hand. But why do cats make this special sound in the first place? Scientists, it turns out, have been asking this question for more than a hundred years — and the answer is stranger and more useful than most people think.

② First, how does a purr happen at all? Unlike a meow, which comes from the throat like a human voice, a purr is made by **fast, regular movements of muscles deep inside the cat's throat**, called the *laryngeal muscles*. These muscles open and close the airway about 25 to 150 times per second, and the rushing air makes the buzzing sound we hear. A purring cat can breathe in **and** breathe out, yet the sound keeps going almost without a pause.

③ For a long time, people thought cats purr only when they are happy. In some ways this is true: cats often purr while eating, being stroked, or resting in the sun on their owner's lap. These are classic "contentment purrs". But careful observation has shown something surprising. Cats also purr in **situations that do not look happy at all** — at the vet, when they are hurt, when they are giving birth, and even when they are dying. A purr, it seems, is not a simple "smile."

④ This has led scientists to a more interesting idea: **the purr is also a self-healing tool**. Research has shown that the frequency of a typical cat's purr is between **20 and 140 hertz** — and frequencies in this range are known, in human medicine, to help bones grow faster, muscles heal, and pain become less strong. One well-known study even used **low-frequency vibration therapy** on astronauts, whose muscles weaken in space. Cats, in other words, may carry a tiny physical therapy clinic inside their own bodies. When an injured cat curls up in a corner and purrs for hours, its body may quietly be repairing itself.

⑤ Purring is also **a way of talking to humans**. Some cats have learned to produce a special kind of purr when they want to be fed. Scientists call this the "**solicitation purr**". It hides a higher-pitched "cry" sound inside the lower-pitched purr. To the human ear, the sound is pleasant but slightly urgent — much like a baby's cry. A hungry cat who purrs like this in the morning often gets a very quick breakfast indeed.

⑥ Interestingly, **not all big cats in the cat family can purr**. Lions and tigers can roar, but they cannot produce a steady continuous purr like a house cat. Cheetahs, however, can purr beautifully, but they cannot roar. The difference lies in the shape of a small bone,

the *hyoid*, in the throat: flexible in roaring cats, stiffer in purring cats. Evolution, it seems, made you choose.

⑦ So the next time your grandmother's old cat sits warm on your knees, close your eyes and listen. That soft motor is many things at once: a sign of trust, a greeting, a quiet hospital for the cat's own tired body, and — very often — a polite and clever request for dinner.

注释： purr 咕噜声；laryngeal muscles 喉部肌肉；frequency 频率；hertz 赫兹（Hz）；solicitation 恳求；evolution 进化

根据文章内容，选择最佳答案。

1. How is the purring sound produced?

A. By the cat's tongue hitting its teeth very quickly.
B. By muscles deep in the throat (the laryngeal muscles) opening and closing the airway 25 — 150 times a second, so that air flowing in and out makes a buzzing sound.
C. Only by the cat breathing in.
D. By the cat swinging its tail.

2. According to Paragraph 3, which of the following is true about when cats purr?

A. Cats only purr when they are completely happy.
B. Cats never purr during painful or sad situations.
C. Cats purr not only when content but also when hurt, at the vet, giving birth, or even when dying — a purr is not a simple "smile".
D. Cats purr mainly to annoy other animals.

3. Why do scientists think a purr might help the cat heal?

A. Because purring warms up the cat's fur.
B. Because the frequency of the purr (20 — 140 Hz) is in a range known to help bones grow faster, muscles heal, and pain decrease.
C. Because purring puts other cats to sleep.
D. Because the purr calls doctors to come.

4. What is special about the "solicitation purr"?

A. It is silent.
B. It is made only by big cats like lions.
C. It hides a higher-pitched "cry" sound — similar to a baby's cry — inside the lower-pitched purr, and cats use it to quickly get food from humans.
D. It is a dangerous warning to other animals.

5. According to Paragraph 6, why can't lions and tigers purr like house cats?

A. They are too busy hunting.
B. The shape of a small throat bone, the hyoid, is different: flexible in roaring cats, stiffer in purring cats — so roaring and steady purring do not come together.
C. They do not want to purr.
D. They only purr underwater.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "fast, regular movements of muscles ... called the laryngeal muscles ... 25 to 150 times per second ... the rushing air makes the buzzing sound"
【解析】喉部肌肉有节律开合气道，气流发声。A、C、D 均错。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第③段 "Cats also purr in situations that do not look happy at all — at the vet, when they are hurt ... a purr, it seems, is not a simple 'smile.'"
【解析】猫在痛苦时也会咕噜。A、B 绝对化、D 曲解原文。故选 C。

3.
【答案】B
【题型】细节理解题
【定位】第④段 "frequency of a typical cat's purr is between 20 and 140 hertz ... help bones grow faster, muscles heal, and pain become less strong"
【解析】该频率具理疗作用。A "取暖"、C "催眠别猫"、D "召唤医生"均无依据。故选 B。

4.
【答案】C
【题型】细节理解题
【定位】第⑤段 "It hides a higher-pitched 'cry' sound inside the lower-pitched purr ... much like a baby's cry. A hungry cat ... often gets a very quick breakfast"
【解析】讨食咕噜内含婴儿啼哭音。A、B、D 均错。故选 C。

5.
【答案】B
【题型】推理判断题
【定位】第⑥段 "The difference lies in the shape of a small bone, the hyoid, in the throat: flexible in roaring cats, stiffer in purring cats."
【解析】舌骨形状决定能吼或能咕噜。A、C、D 均非事实。故选 B。